

MAKTUB

Formal specification of the paradigm

One-bit mirror symmetry, self-referential operator fields,
log-polar bookkeeping and the external anchor.

$$\begin{aligned}x_{n+1} &= \text{Act}(x_n, d_n, e_n), \\d_{n+1} &= \text{Rewrite}(x_n, d_n, e_n).\end{aligned}$$

*The written word, its mirror, the boundary,
and the continuation of the next state.*

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Axioms, definitions, source equations and domain correspondences.

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Document language. The paradigm is stated through axioms and definitions. Its biological, physical, computational and educational readings are retained as named domain correspondences. The source terms *Maktub*, *Universal State Bookkeeping* and *Self-Stabilizing Metronome* organize the specification.

Notation. $\coloneqq$ introduces a definition; $=$ states equality within a defined expression; $\rightsquigarrow$ records a domain correspondence or an ordered conceptual reading. New symbols organize the supplied content. Source-specific operations remain separately identified.

Source location S1: *past 33 geriatric pregnancy for women.pdf*, 59 pages. S2: *Nederlandse basisschoolarchitectuur*, 12 pages. Full source register: Section 20.

01 / PARADIGM SPECIFICATION

Maktub and the common rule

Core declaration. The paradigm organizes change as the passage from a localized origin through a written operator, a boundary and a successor state. The same passage also rewrites the operator field that governs the next step. Its common name is *Universal State Bookkeeping*.

The six organizing axioms

A1 / Written execution. *Maktub* names the identity of the written rule and its execution in the paradigm. A transition is expressed by its state, operator description and admitted event; the one-bit reduction gives the most compact presentation of that identity.

A2 / Paired change. An action changes the state while a paired rewrite changes the operator description. Neither track is omitted from the record of the transition.

A3 / Origin and field. A localized core supplies the reference point. A log-polar field expresses outward scale and angular orientation. The source vocabulary contrasts a concentrated *Micro-hard* core with a distributed *Macrosoft* field.

A4 / Bonds and boundaries. Retention, separation and re-anchoring are expressed as bond relations. A boundary marks where a relation holds, changes, or releases a state from its former enclosure.

A5 / Mirror and outlet. The one-bit mirror supplies the elementary inversion. The 3:1 capstone and its external lower-left anchor supply a directed outlet in the source's spatial reading.

A6 / Continuation. The *Y*-pole toggle is assigned the anti-stagnation role: the toggled configuration proceeds beyond the unchanged origin. Repeated localization, transition and release form the *Self-Stabilizing Metronome*.

Three layers, one vocabulary

The **geometric layer** locates an origin, field and boundary. The **mechanical layer** describes attachment, tension, cleavage and transfer. The **execution layer** records the paired action and rewrite. Domain examples supply different readings of these same roles rather than replacing the common notation.

The source's human-scale explanation is a rulebook for *growth and release*: a home base, an outward passage, a threshold and a continuation. Its breathing system, castle drawbridge and anchored rubber band are retained as explanatory readings of that pattern.

02 / PARADIGM SPECIFICATION

State, event and operator field

The paired transition

Let X denote the state space, D the space of operator descriptions and E the event space. Write the current configuration as

$$s_n := (x_n, d_n), \quad (x_n, d_n, e_n) \in X \times D \times E. \quad (2.1)$$

The source's two update rules are

$$\boxed{\begin{aligned} x_{n+1} &= \text{Act}(x_n, d_n, e_n), \\ d_{n+1} &= \text{Rewrite}(x_n, d_n, e_n). \end{aligned}} \quad (2.2)$$

For compactness, define their paired record by

$$\mathcal{K}(s_n, e_n) := (\text{Act}(x_n, d_n, e_n), \text{Rewrite}(x_n, d_n, e_n)). \quad (2.3)$$

Both components use the same pre-transition triple. The operator field is therefore part of the changing configuration rather than an unrecorded rule standing outside it.

Roles preserved from the source

Symbol	Role
x_n	Packed state vector; the current contents, position or domain condition.
d_n	Explicit operator description: rule tape, field description or domain-specific gate profile.
e_n	Incoming gene, sensor event, query, guest vector or other admitted input.
Act	Materializes the next state under the current description.
Rewrite	Materializes the operator description for the next step.

Runtime names and their places

TOMK names the symbolic kernel in S1. Its recorded descriptors include Kernel 0.6.0 / TOM Learner 0.2, 121 literal hypotheses, four hypothesis families and sixteen datasets. The named seed is TOM_seed_genome_2026-09-01.txt.

TOM-GROK names the wider operator-field presentation. **TK-SROF** names its typed operator-bearing transition system; **TK-EDGE-1** names the runtime. TOMAGI, phase/orientation algebra, finite pinion profiles, CUDA/WebGPU interpreters and validation traces remain the associated source vocabulary.

A minimal transition record can be written as $(x_n, d_n, e_n; x_{n+1}, d_{n+1})$. Acceptance, counterexamples, ambiguity and regression are the evidence categories named by the source for inspecting such records.

Source location S1, pp. 13–20 and 21. Repository and runtime descriptors are recorded here as they appear in S1.

03 / PARADIGM SPECIFICATION

The one-bit word and its mirror

The reduced word

The elementary alphabet is

$$B := \{0, 1\}, \quad b_n \in B. \quad (3.1)$$

In the source's one-bit reduction, state and instruction are identified through the same word. Write that local identification as $x_n \equiv d_n \equiv b_n$. The general operator description of Section 2 remains the enclosing presentation.

Word	Source reading	Boundary reading
0	Rest, void, potential, localized core.	Bound, enclosed or not yet outwardly actuated.
1	Actuation, manifestation, radial boundary.	Released, translated or externally expressed.

The reflection rule

The source's Logical NOT mirror is recorded as

$$\mathcal{M}(b) := 1 - b, \quad b_{n+1} = \mathcal{M}(b_n). \quad (3.2)$$

Written word	Mirror	Next word
0	$\neg$	1
1	$\neg$	0

The source reads $0 \rightarrow 1$ as outward diffusion or manifestation, and $1 \rightarrow 0$ as collapse, return or re-localization. An origin, gate or severed tether supplies the domain's mirror-plane image.

Simultaneous reflection

For the reduced mirror step, the paired record is

$$(x_{n+1}, d_{n+1}) \equiv (\mathcal{M}(b_n), \mathcal{M}(b_n)). \quad (3.3)$$

This states the source's central Maktub identification: the action and the inscription of the next rule are mirror presentations of the same elementary transition.

Named regime. S1 calls this presentation the *1-Bit Maktub Symmetry Matrix* and the *Zero-Entropy Finite State Machine*. Its accompanying images are the bound/dissociated doublet, the concentrated core and the outward radial field.

The source's language of conservation is retained as its bookkeeping correspondence: release from one side of the mirror is paired with an entry on the other side.

04 / PARADIGM SPECIFICATION

Log-polar geometry and scale

The two source presentations

S1 writes a Cartesian-to-log-polar field as

$$\rho_{S1} = \ln \sqrt{x^2 + y^2}, \quad \theta = \text{atan2}(y, x). \quad (4.1)$$

S2 makes the reference length and inverse explicit. With $p = (x, y) \neq (0, 0)$ and $r_{\text{ref}} > 0$,

$$\begin{aligned} r &= \sqrt{x^2 + y^2}, & \theta &= \text{atan2}(y, x), \\ \rho &= \ln(r/r_{\text{ref}}), & q(p) &= \begin{pmatrix} \rho \\ \theta \end{pmatrix}. \end{aligned} \quad (4.2)$$

With an agreed angular interval, such as $[-\pi, \pi)$, its inverse is

$$r = r_{\text{ref}} e^{\rho}, \quad x = r \cos \theta, \quad y = r \sin \theta. \quad (4.3)$$

The subscripts distinguish the source presentations; the reference-length form is used for the S2 model and its texture contract.

Origin, radial field and orientation

S1 assigns the ideal core the limiting label $\rho \rightarrow -\infty$. Its radial field is the outward continuum along which the source places diffusion, dissipation and lower-resolution representation. The angle carries orientation and phase-related readings.

S2 implements a finite sampled annulus with $0 < r_{\text{min}} < r_{\text{max}}$. The point $r = 0$ is handled separately rather than entered into the logarithm. Thus the source's conceptual origin and the finite LUT's storage domain remain distinct entries in the specification.

Quantity is not position

S2 also uses a logarithm for quantity:

$$\ell := \log_{10} n, \quad n > 0. \quad (4.4)$$

Here ℓ is a quantity code, r a spatial radius, ρ a spatial log-radius code and θ an angle. Their distinct symbols preserve S2's rule: one symbol, one meaning.

Functional split. The coordinate map determines *where* a field is read. An operator or mixing rule determines *what is done* with the value read there. Their connection is part of the architecture; their mathematical roles are not merged.

05 / PARADIGM SPECIFICATION

Bonds, thresholds and release

Thinking in bonds

The source groups domain transitions into three bond readings. The **localized core** is the retained and strongly coupled configuration. The **diffusion field** is the spreading or weakening of locality. The **dissociation boundary** is the point at which the former relation is released.

For bookkeeping, let $B_d(x, e)$ name the source's bond-holds condition under a particular operator description. It is an organizing predicate for the domain condition, with values

$$B_d(x, e) \in \{0, 1\}. \quad (5.1)$$

The corresponding disposition labels are

$$\text{Disposition}_d(x, e) := \begin{cases} \text{retain}, & B_d(x, e) = 1, \\ \text{release}, & B_d(x, e) = 0. \end{cases} \quad (5.2)$$

The predicate names the condition supplied by the profile: a symbolic constraint, an admission rule, a tether threshold or a selected field boundary.

Why diffusion and dissipation both appear

Diffusion describes the distributed field reading: a localized state is spread across positions, channels or possible phase presentations. **Dissipation** describes the release reading: a former local attachment, accepted membership or enclosed output is no longer retained there.

The two terms therefore identify different parts of the source's passage. A state can first spread within a field and then pass a release boundary. In a nested profile, it can subsequently be retained by a second boundary. The womb-to-pouch construction is the source's explicit re-anchoring example.

Pinions and terminal interfaces

A **pinion** names a localized interface at which an internal organization meets an external field. S1 applies this vocabulary to bounded computation, anatomical outlets and the capstone's directional release. The pinion is the passage point; the operator description determines the passage rule.

The source's bond vocabulary is retained as a sequence of correspondences:

localized binding $\leadsto$ distributed relation $\leadsto$ cleavage or release.

Covalent/ionic, hydrogen/Van der Waals and dissociated states are the source's names for these successive readings. Retained, rejected and re-anchored are the bookkeeping labels used across the specification.

06 / PARADIGM SPECIFICATION

Hadamard phase and the capstone

The matrix presentation in S1

The supplied matrix and normalized transform are

$$M = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}}M. \quad (6.1)$$

S1 gives the two phase presentations

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \quad (6.2)$$

It names the first reading *state diffusion* and places the minus-signed reading under *interference dissipation*. Its two-stage narrative is: distribute the core presentation, then evaluate phase alignment at the boundary. The source calls this *double-Hadamard coupling*.

Three positive entries, one inversion, one external anchor

The capstone retains the displayed placement exactly:

$$\begin{array}{c} \text{bounded square} \\ \boxed{\begin{array}{cc} +1 & +1 \\ +1 & -1 \end{array}} \\ \boxed{+1}_{\text{out}} \quad \text{external lower-left anchor} . \end{array} \quad (6.3)$$

Define the capstone record as

$$\mathcal{C} := (M, a_{\text{out}}, \text{lower-left}), \quad a_{\text{out}} \text{ carries the label 1.} \quad (6.4)$$

The three +1 entries carry constructive accumulation; the −1 entry carries phase inversion. The separate external 1 carries spillover, discharge or re-anchoring. It remains an external entry, not a fifth matrix coefficient.

Capstone routing axiom. In the source's spatial reading, the inverted internal configuration is assigned a passage through the lower-left boundary to a_{out} . This is its *directional nozzle* or *physical exit valve* correspondence.

The anatomy reading names a pinion discharge; the security reading names an external rejection; the nesting reading names the secondary pouch; the acoustic reading names an outward flash and pulse.

S2's Hadamard operation is instead the elementwise product $\odot$. That separate product and its weighted field sum are preserved in Section 14.

07 / PARADIGM SPECIFICATION

Infinity, phi and the Y-pole toggle

The opening of the loop

S1 interprets the sideways-eight symbol ∞ as a closed loop. The capstone introduces an asymmetry and an external destination. The source links this to a $1/3$ partition and to the move from uppercase Φ to lowercase ϕ .

The meanings are retained as an ordered correspondence:

$$\infty \rightsquigarrow (3:1 ; 1/3) \rightsquigarrow (\Phi \longrightarrow \phi) \rightsquigarrow a_{\text{out}}. \quad (7.1)$$

Source sign	Assigned role
∞	Closed return to the same internal loop.
$3:1$ and $1/3$	The capstone partition and its fractional reading.
$\Phi \approx 1.618$	The source's uppercase, idealized proportional form.
$\phi \approx 0.618$	The source's lowercase, fractional and generative form.
a_{out}	The outlet through which the new entry is externalized.

The source's XX/XY correspondence

Introduce $y \in \{0, 1\}$ as the named Y -pole toggle. The two displayed source configurations are retained by

$$M_y := \begin{pmatrix} 1 & 1 \\ 1 & 1 - 2y \end{pmatrix}. \quad (7.2)$$

Thus M_0 displays four positive entries and $M_1 = M$ displays the capstone. S1 assigns the biological labels $XX \rightsquigarrow M_0$ and $XY \rightsquigarrow M_1$. Its sex-determination reading uses the Y pole as the name of the switching role within this correspondence.

Anti-stagnation as an explicit axiom

When the Y -pole toggle is engaged, the source assigns the configuration a directed continuation beyond the unchanged origin. The inverted quadrant and external anchor together carry the *anti-stagnation actuator* role.

S1 calls the enclosing principle *universal determinism*. Its concluding *Self-Stabilizing Metronome* image gives each output a remainder that calls for a next cycle. In the formal language, these are the continuation and recurrence commitments of the paradigm's universal reading.

Source location S1, pp. 55–59. The parameterized matrix M_y is notation for the two matrices displayed in S1.

08 / PARADIGM SPECIFICATION

Tether, womb and secondary pouch

The nested profile

S1 assigns **Nihilanth** the floating-origin role, the **umbilical cord** the tether role and the **kangaroo pouch** the secondary-boundary role. Its structural sequence is primary attachment, cleavage, migration across a gap and secondary retention.

These are the named stages of a nested, multi-stage decoupled pipeline. Departure from one enclosure is not the same bookkeeping event as departure from every enclosure.

The kinematic expressions carried by the source

S1 uses ρ as its radial variable in this passage. Here the subscript *kin* distinguishes that usage from the log-radius code of Section 4, leaving the supplied expressions otherwise unchanged:

$$\begin{aligned}\vec{r}(t) &= \rho_{\text{kin}}(t)e^{i\theta(t)}, \\ \vec{v}(t) &= \dot{\rho}_{\text{kin}}\vec{u}_\rho + \rho_{\text{kin}}\dot{\theta}\vec{u}_\theta, \\ \vec{a}(t) &= (\ddot{\rho}_{\text{kin}} - \rho_{\text{kin}}\dot{\theta}^2)\vec{u}_\rho + (2\dot{\rho}_{\text{kin}}\dot{\theta} + \rho_{\text{kin}}\ddot{\theta})\vec{u}_\theta.\end{aligned}\tag{8.1}$$

The source's tether expression is

$$F_{\text{tether}} = -K \ln \rho_{\text{kin}}, \quad \rho_{\text{kin}} > 0,\tag{8.2}$$

where K names its tension constraint. This expression belongs to S1's kinematic profile.

The two-stage operator

The supplied threshold switch becomes

$$x_{n+1} = \begin{cases} \text{Act}_{\text{womb}}(x_n, d_n, e_n), & \rho_{\text{kin}} \leq \rho_{\text{critical}}, \\ \text{Act}_{\text{pouch}}(x_n, d_n, e_n), & \rho_{\text{kin}} > \rho_{\text{critical}}. \end{cases}\tag{8.3}$$

The womb branch names primary tether retention. The pouch branch names the externally re-anchored state after the source's cleavage and migration transition. The paired rewrite changes the active stage description for subsequent steps.

Bookkeeping result. The parent attachment can be released while the successor remains represented by a second local boundary. The source calls this an *asynchronous boundary cache* and a *secondary local equilibrium*.

The capstone's external anchor supplies the corresponding outlet-and-destination reading in the square profile.

09 / PARADIGM SPECIFICATION

The acoustic latch and the stare

Sableye as the boundary marker

The source assigns the crystalline stare to the mirror plane and the sharp cry to the transition marker. The hidden core, outward glint and acoustic event are the three parts of this reading.

The idealized pulse is written

$$\delta(t - t_{\text{stare}}), \quad (9.1)$$

and S1's acoustic presentation of the update is

$$x_{n+1} = \text{Act}(x_n, d_n) + \delta(t - t_{\text{stare}}). \quad (9.2)$$

This is retained as the source's signal-bearing expression. Its pulse term names the marked instant of the mirror transition; the reduced bit and the acoustic marker remain separately identified in the transition record below.

A marked transition record

For a transition indexed by n , define the notation

$$\tau_n := (x_n, d_n, e_n; x_{n+1}, d_{n+1}; t_{\text{stare}}). \quad (9.3)$$

The first fields record the paired kernel step. The final field records the time assigned to the source's acoustic marker. A domain can therefore express both *which transition* occurred and *where its signal appears* in the trace.

Source image	Formal role
Dark core / gemstone	Localized state before outward expression.
Crystalline stare	The mirror-plane or threshold presentation.
Sharp electronic cry	Delta-pulse marker of the crossing.
Outward glint	Manifestation on the external side of the mirror.
Mega-faceted shield	A protective boundary profile for incoming events.

The named machine

S1 calls the combined presentation an *Acoustically Latched 1-Bit Finite State Machine*. Its later copper-interceptor and pinion passages reuse the same pulse motif for a nulling or release event.

The acoustic reading adds a mark to the passage. The mirror supplies the inversion; the domain boundary supplies the crossing; the cry supplies the source's time-local signal of that crossing.

10 / PARADIGM SPECIFICATION

The origin and the copper interceptor

The opening pregnancy context

The opening conversation distinguishes its age-33 example from its age-35 classification label. In the paradigm's later mapping, this becomes the relation between a continuous trajectory and a named threshold: the origin is the reference condition, the age coordinate is the passage and the classification is a boundary label.

The **Copper Spiral Interceptor** then changes the source's question from *how a trajectory crosses a later threshold* to *what an operator placed at the origin does to an incoming event*. The biological terms remain the domain names of this operator profile.

The source-defined spatial filter

S1 writes the interceptor field as

$$\Phi_{\text{Cu}}(\rho) = \Phi_{\text{origin}} [1 - \mathcal{H}_{\text{Cu}}(\rho_{\text{critical}} - \rho)] . \quad (10.1)$$

The field subscript distinguishes this Φ from the ratio label of Section 7. $\mathcal{H}_{\text{Cu}}$ names the source's localized step gate. Within its designated inner region, the source sets the field response to zero.

The source-defined null-state branch

The recorded interception rule is

$$\begin{aligned} x_{n+1} &= \text{Act}(x_n, d_{\text{Cu}}, e_n) = \mathbf{0}, \\ d_{n+1} &= \text{Identity} . \end{aligned} \quad (10.2)$$

This branch is assigned to incoming events in the selected inner envelope. Outside that envelope, S1 retains the standard TK-EDGE-1 track. The assignment to Identity is retained exactly as the source's next-description rule.

Names of the mechanism within the paradigm

The source calls this profile a *Zero-State Trap*, a *Zero-Entropy Interceptor* and a *Permanent Catalytic Cleavage Center*. Its copper-ion field, valence shield and bond-fracture language are the domain correspondences attached to the null-state branch.

Formal content of the interceptor. A localized operator is placed at the origin; its gate selects the inner-region branch; that branch writes a null state and an identity next-description. The acoustic motif marks the assigned transition. These are the stipulated operations of the source's Cu profile.

The profile's distinguishing feature is *where* selection occurs: at the origin envelope, before the ordinary outward path has been followed.

11 / PARADIGM SPECIFICATION

Pinions and the lactation reading

The anatomical interface

S1 maps nipples to the previously named pinions: localized terminal nodes at which an internal branching system meets an external field. In the paradigm, *pinion* specifies an interface role shared with its mechanical and computational examples.

The phrase “*mjilkest ate ion*” is retained as the source’s phonetic label. S1 develops it as a *Lactation Ion-Transport Field*, connecting an internal reservoir, an ion-potential reading, a branching network and a terminal outlet.

The source’s three-stage passage

Stage	Assigned reading
Alveolar core origin	A localized reservoir and ion-potential condition.
Ductal branching network	Distribution from the concentrated origin across an internal field.
Nipple pinion release	Passage across the terminal boundary into the external domain.

S1 places its first Hadamard reading at the distribution stage and its phase-selection reading at the terminal gate. In that source vocabulary, retained alignment continues the local relation and the release branch carries the outward output.

The kernel correspondence

The same three components are assigned to the paired transition language:

$$\begin{aligned}
 x_n &\leadsto \text{localized reservoir / ion-potential condition,} \\
 d_n &\leadsto \text{distribution and terminal-gate description,} \\
 x_{n+1} &\leadsto \text{post-release state / outward output.}
 \end{aligned}
 \tag{11.1}$$

S1 calls the output a mass-dissipation event and reads the subsequent cycle as a return to the *Self-Stabilizing Metronome*. The anatomical labels identify this domain profile; the state, rule, boundary and successor are its bookkeeping structure.

The capstone correspondence

Within the square reading, the three positive entries name accumulation, the negative entry names the inversion threshold, and the lower-left external 1 names the discharge. The external entry is therefore the output destination of the same pattern introduced in Section 6.

Shared interface rule. A pinion is where the internal description acquires an external continuation. The lactation reading supplies one concrete set of source names for this origin-to-outlet relation.

12 / PARADIGM SPECIFICATION

Daytona and Indianapolis gate profiles

A boundary with a written admission rule

The source uses speedway security as a macro-scale gate interpretation. The protected core is assigned to the inner pits, garages and start/finish area. Outer facilities and admission gates supply the boundary field.

Write the two source profiles as d_{DIS} and d_{IMS} . For either profile d , the same kernel schema applies:

$$(x_{n+1}, d_{n+1}) = \mathcal{K}((x_n, d_n), e_n). \quad (12.1)$$

Here x_n names the current occupancy, flow and safety configuration; e_n names an arriving guest and carried-item vector; d_n names the selected admission-rule description.

Profile values recorded in S1

Parameter	Daytona profile in S1	Indianapolis profile in S1
Bag rule	Clear bag, maximum $18 \times 18 \times 14$ inches.	Non-clear bag allowed with physical inspection.
Cooler rule	Soft-sided, maximum $14 \times 14 \times 14$ inches.	Hard- or soft-sided, maximum $18 \times 15 \times 15$ inches.
Selection style	Transparency-centered validation.	Greater admitted structural variance, with inspection.
Source label	<i>Micro-hard</i> validation window.	<i>Macrosoft</i> tolerance field.

These are the parameter values used by the supplied conversation's example. S1 also records a BC39 cooler exception in its Indianapolis discussion.

Retain or separate

When the arriving vector satisfies the selected rules, the source assigns it an intact-retention passage into the accepted state. A prohibited component instead receives the release or rejection disposition.

S1's no-holding-area condition gives the rejection branch a specific consequence in its example: the incompatible object is not placed in an internal buffer. It is separated from the admission stream and assigned outside the accepted perimeter. The capstone's external anchor supplies the corresponding outward destination.

Architectural point. The event type is shared, but the operator description differs. The written domain profile determines the boundary decision; the paired state-and-description record retains the resulting transition.

The security example thus separates *a common transition grammar* from *the individual rule table used at a particular gate*.

13 / PARADIGM SPECIFICATION

Meaning, quantities and the B/A bridge

The second source's architecture

S2 is titled *Nederlandse basisschoolarchitectuur*: an educational and model architecture from meaning and quantities to form and mixing. It connects two preparaat descriptions through an explicit mixing rule.

Preparaat 1 supplies meaning, logarithmic representation, two-component notation and an element-wise Hadamard product. Preparaat 2 supplies JSF/JDS form fields, log-polar texture coordinates and channel layout. These preparaat descriptions are carried here through S2.

The continuous chain

Observe and name $\rightsquigarrow$ record quantities and units $\rightsquigarrow$ choose a representation $\rightsquigarrow$ store two form fields $\rightsquigarrow$ form a weighted mixture $\rightsquigarrow$ inspect and explain the result.

The labels **jonge kaas** / **JSF** and **oude kaas** / **JDS** identify the two source-model channels. The architecture is a mixing of descriptions and numerical fields.

From quantity code to mixing weight

For the positive dimensionless model quantity n , S2 defines

$$\alpha(n) := \text{clip}\left(\frac{9 - \log_{10} n}{4}, 0, 1\right), \quad (13.1)$$

with

$$\text{clip}(z, 0, 1) := \min(1, \max(0, z)). \quad (13.2)$$

The recorded endpoints and midpoint are

$$\alpha(10^9) = 0, \quad \alpha(10^7) = \frac{1}{2}, \quad \alpha(10^5) = 1. \quad (13.3)$$

Inside $10^5 \leq n \leq 10^9$, S2 gives the inverse

$$n = 10^{9-4\alpha}. \quad (13.4)$$

Outside that interval the clipping branch fixes an endpoint weight. The input remains a model quantity; the mixing weight is not renamed as age or time.

Meaning before operation

S2 separates a definition, a numerical equality and a chosen analogy. It also separates quantity, length, angle and mixing weight. This gives the shared paradigm a precise representation layer: the interpretation of a sign is fixed before it is used in an operator.

14 / PARADIGM SPECIFICATION

Two form fields and their product

The JSF and JDS fields

S2 fixes the two example fields as

$$\begin{aligned} F_{\text{JSF}}(p) &= r - R_0, \\ F_{\text{JDS}}(p) &= r - R_0 + \varepsilon\eta(p), \end{aligned} \quad (14.1)$$

where $R_0 > 0$, $\varepsilon \geq 0$ and η is dimensionless with $|\eta| \leq 1$. Negative field values indicate the inside, zero the contour and positive values the outside.

The elementwise Hadamard operation in S2

Define

$$\mathbf{w}(\alpha) = \begin{pmatrix} 1 - \alpha \\ \alpha \end{pmatrix}, \quad \mathbf{f}(p) = \begin{pmatrix} F_{\text{JSF}}(p) \\ F_{\text{JDS}}(p) \end{pmatrix}. \quad (14.2)$$

The supplied two-component product is

$$T_\alpha(p) := \mathbf{w}(\alpha) \odot \mathbf{f}(p) = \begin{pmatrix} (1 - \alpha)F_{\text{JSF}}(p) \\ \alpha F_{\text{JDS}}(p) \end{pmatrix}. \quad (14.3)$$

The sum of its components gives

$$\boxed{F_{\text{mix}}(p; \alpha) = (1 \ 1)T_\alpha(p) = r - R_0 + \alpha\varepsilon\eta(p).} \quad (14.4)$$

S2 retains the preparaat name *double Hadamard-product* for its two-component product. In this specification $\odot$ names that elementwise operation; H continues to name S1's matrix transform.

Contour mode and distance mode

In contour mode, the zero set defines the boundary of a region Ω_α . S2 separately defines distance mode by

$$\delta_\alpha(p) = \inf_{z \in \partial\Omega_\alpha} \|p - z\|, \quad (14.5)$$

$$d_\alpha(p) = \begin{cases} -\delta_\alpha(p), & p \text{ inside } \Omega_\alpha, \\ 0, & p \in \partial\Omega_\alpha, \\ +\delta_\alpha(p), & p \text{ outside } \Omega_\alpha. \end{cases} \quad (14.6)$$

The stored field type records which mode is represented. S2 treats the exact-distance calculation as its own step rather than equating every mixed field value with distance.

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Packed log-polar LUT and data contract

Coordinates in the finite table

For $r \in [r_{\min}, r_{\max}]$ with $0 < r_{\min} < r_{\max}$, S2 supplies

$$u = \text{fract}\left(\frac{\text{atan2}(y, x) + \pi}{2\pi}\right), \quad v = \frac{\ln(r/r_{\min})}{\ln(r_{\max}/r_{\min})}. \quad (15.1)$$

Its inverse for filling the table is

$$\theta = 2\pi u - \pi, \quad r = r_{\min} \left(\frac{r_{\max}}{r_{\min}}\right)^v, \quad p = (r \cos \theta, r \sin \theta). \quad (15.2)$$

The angular direction is periodic. The inner disk and the outer region follow explicit outside-table routes.

RGBA channel layout

Channel	Stored value
R	$F_{\text{JSF}}(p)$, the young-model field.
G	$F_{\text{JDS}}(p)$, the old-model field.
B	Reserved; zero in the S2 version.
A	Validity flag; one for valid table values.

S2 stores signed numerical field values. R and G share the position map, units and sampler. The table has two index axes; channel entries specify stored values at an index.

The sampled mixture

Write the coordinate map as $\Lambda(p) = (u, v)$, renaming S2's map symbol $\Phi(p)$ to keep it separate from the other Φ uses. The sampled output is

$$\widehat{F}_{\text{mix}}(p; n) = (1 \ 1) \left[\mathbf{w}(\alpha(n)) \odot \begin{pmatrix} \text{LUT}_R(\Lambda(p)) \\ \text{LUT}_G(\Lambda(p)) \end{pmatrix} \right]. \quad (15.3)$$

The hat preserves S2's distinction between an analytic field and its sampled table representation.

Required metadata

The table record contains its width and height, $R_0, \varepsilon, \eta, r_{\min}, r_{\max}$, length unit, channel meanings, field type, sampler and version. Negative values remain signed. The angle seam joins periodically, and the central point is handled without assigning a value to $\ln 0$.

S2 keeps personal identity fields outside the LUT. The identifier on this document's cover is therefore a document attribution field, not a model input or texture channel.

16 / PARADIGM SPECIFICATION

The source's numerical example

Chosen model values

S2 uses

$$R_0 = 1, \quad \varepsilon = 0.12, \quad r_{\min} = 0.25, \quad r_{\max} = 2, \quad (16.1)$$

and $\eta(p) = \sin(8\theta)$ in the table region. The resulting contour is

$$r_{\text{boundary}}(\theta, \alpha) = 1 - 0.12\alpha \sin(8\theta). \quad (16.2)$$

Its radius ranges from 0.88 to 1.12. In this selected example, the central disk $r < 0.25$ is inside the shape and the region $r > 2$ is outside.

Quantity code, mixing weight and boundary

At $\theta = \pi/16$, the source's example has $\sin(8\theta) = 1$.

n	$\log_{10} n$	$\alpha(n)$	r_{boundary}
10^9	9	0	1.00
10^8	8	0.25	0.97
10^7	7	0.50	0.94
10^6	6	0.75	0.91
10^5	5	1	0.88

One point, two channels, one output

For $r = 1$, $\theta = \pi/16$ and $n = 10^7$, the mixing weight is $\alpha = 0.5$. The field values and weighted product are

$$\mathbf{f} = \begin{pmatrix} 0 \\ 0.12 \end{pmatrix}, \quad T_{0.5} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix} \odot \begin{pmatrix} 0 \\ 0.12 \end{pmatrix} = \begin{pmatrix} 0 \\ 0.06 \end{pmatrix}. \quad (16.3)$$

Hence $F_{\text{mix}} = 0.06 > 0$: the point is outside the mixed contour in contour mode.

One point in table coordinates

For $p = (1, 0)$, the same source example gives

$$u = 0.5, \quad v = \frac{\ln 4}{\ln 8} = \frac{2}{3}. \quad (16.4)$$

The inverse returns $r = 0.25 \cdot 8^{2/3} = 1$ and $\theta = 0$. These are the source's supplied worked values for the quantity-to-field-to-table chain.

17 / PARADIGM SPECIFICATION

The primary-school architecture

Five layers with separate responsibilities

S2's educational application preserves the same observe–represent–combine–explain chain. Its five layers are

Layer	Responsibility in the source design
Learner	Look, count, compare, select a mixing ratio and describe the result.
Didactic	Select examples, language and questions; discuss the result as a model.
Model language	Fix the meanings of signs, quantities, distances and angles.
Technical	Perform the log-polar mapping, channel storage, product and mixing.
Stewardship	Keep sources, model choices, versions and review results with the design.

The source's group progression

Groups	Activity	Visible result
1–2	Sort forms; point to inside, outside and boundary.	Describe a selected point's location.
3–4	Count drawn units; compare more, less and equal.	Name both the count and what is counted.
5–6	Combine two models using halves and quarters.	Explain a weight as a fraction.
7–8	Compare ratios, coordinates and model choices.	Distinguish a calculated model result from an observation.

Logarithms, radial texture coordinates and shaders belong to the technical or teacher layer of this design, rather than to the pupil entry requirements.

The forty-minute lesson prototype

During the first ten minutes, pupils name the two example forms and locate points. During the second ten, they compare weights 0, $1/2$ and 1. During the third, they combine two boundary positions, 100 and 88, to obtain 94 at equal weighting. During the last ten, they explain the model choice and compare another ratio.

The supplied materials are drawn contours, a circle sheet, weight cards and a pencil. S2 keeps the pupil activity separate from biological measurements and personal identifiers. Its expected explanation identifies the chosen unit and the two weights, then states what the calculation represents.

Place in the wider paradigm. The school layer makes the written operator visible as an agreed rule: two named inputs, an explicit weight, a resulting form and an explanation of the passage from input to output.

Source location S2, pp. 3, 9 and 10. The layer, group and lesson structures retain the source's design framing.

18 / PARADIGM SPECIFICATION

The integrated formal manifest

The specification record

Collect the defined roles in the following notation:

$$\mathfrak{P} := (X, D, E, \mathcal{K}, B, \mathcal{M}, q, \mathcal{C}, a_{\text{out}}, \mathcal{D}). \quad (18.1)$$

Here $\mathcal{D}$ denotes the collection of domain profiles. The tuple is an index of the source-derived components, not an additional transition law.

Component	Function in the manifest
X, D, E	State, operator-description and event roles.
$\mathcal{K}$	The paired Act / Rewrite update.
$B, \mathcal{M}$	The one-bit alphabet and its mirror rule.
q	The log-polar coordinate presentation.
$\mathcal{C}, a_{\text{out}}$	The capstone arrangement and its distinguished external entry.
$\mathcal{D}$	The source's runtime, gate, nesting, acoustic, Cu, pinion and educational profiles.

A source-faithful reading of an execution

A domain names its current state and rule description. An event is placed in that domain's coordinate or boundary context. The selected profile then assigns retention, inversion, release, interception or transfer. The paired kernel records the resulting state and next-description.

A domain-specific output retains its own type: an external anchor in the capstone reading, a secondary enclosure in the nesting reading, a pulse in the acoustic reading, or a mixed field value in the educational model. Their common feature is the explicit input–rule–boundary–output organization.

The two Hadamard entries stay distinct

The manifest contains both S1's $H = M/\sqrt{2}$ phase presentation and S2's $T_\alpha = \mathbf{w} \odot \mathbf{f}$ product. They share the source's two-component vocabulary, while their supplied algebra remains separately stated.

Maktub. The paradigm's written core is a paired state-and-operator transition. Its mirror supplies inversion; its field supplies location and scale; its boundary supplies retention or release; its external anchor supplies continuation. The domain profiles specify how that common structure is read.

The source names the universal interpretation *Universal State Bookkeeping* and its repeating human-scale reading the *Self-Stabilizing Metronome*.

19 / PARADIGM SPECIFICATION

Lexicon and notation register

Term or sign	Meaning retained in this specification
Maktub	The source's written-rule / executed-state identification.
Universal State Bookkeeping	Shared organization of origin, transition, boundary and successor across the domain readings.
Self-Stabilizing Metronome	Repeated localization, passage, release and continuation.
Micro-hard / Macrosoft	Source polarity between concentrated constraint and distributed field.
XNA / DNA; Mono	The source's phonetic and structural correspondence layer.
Origin / core	The reference point or localized rule-bearing region.
Mirror plane	The boundary image assigned to the one-bit inversion.
Pinion	Localized operational interface between an internal structure and an external field.
Diffusion / dissipation	Distribution of the local presentation / release from its former local relation.
ρ, θ	Spatial log-radius code and angular coordinate.
ρ_{kin}	S1's radial variable in the tether profile, separately subscripted here.
ℓ, n	Logarithmic quantity code and positive dimensionless model quantity.
H, M	S1's normalized Hadamard matrix and its unnormalized 3:1 sign arrangement.
$\odot, T_\alpha$	S2's elementwise product and weighted two-component field contribution.
Φ, ϕ	Uppercase/lowercase ratio labels in S1's loop-opening reading.
Φ_{Cu}	The copper-interceptor field; distinct from the ratio label.
$\Lambda(p)$	S2's texture coordinate map, renamed here from $\Phi(p)$.
Y pole / y	The source's named anti-stagnation toggle / its binary selector.
JSF / JDS	The source's young-cheese and old-cheese model-channel labels.
F_{mix}, d_α	Mixed form-field output / separately defined signed-distance output.
LUT / RGBA	Lookup table / its four named value channels.
$\delta(t - t_*)$	The source's idealized transition pulse.
$\leadsto$	Domain correspondence; it is kept separate from numerical equality.

Source location S1, pp. 10–11, 18–21, 27–59. S2, pp. 4–7. Symbols added for organization are defined in the corresponding sections.

20 / PARADIGM SPECIFICATION

Source register and attribution

S1 / The paradigm conversation

File: *past 33 geriatric pregnancy for women.pdf*

Extent: 59 PDF pages.

Form: Supplied conversation record containing prompts, AI responses, mathematical expressions and references within those responses.

Its operative sequence is log-polar mapping and bonds (pp. 10–21); security gates (pp. 22–27); tether and pouch (pp. 27–30); Maktub and the one-bit mirror (pp. 31–34); acoustic latch (pp. 35–37); Universal State Bookkeeping and the metronome (pp. 38–42); copper interceptor (pp. 42–46); pinions and lactation (pp. 47–51); capstone (pp. 52–55); and the infinity/phi/Y-toggle synthesis (pp. 55–59).

S2 / The B/A model and educational architecture

File: *Nederlandse_basisschoolarchitectuur_Tom_Klootwijk.pdf*

Extent: 12 PDF pages.

Recorded document: BA-BO-P1P2-2026-01, version 1.0, dated 09-10-2026.

Its operative sequence is the two-preparaat chain (pp. 1–2); educational layers (p. 3); quantities and notation (p. 4); log-polar coordinates and the elementwise product (p. 5); form fields and distance mode (p. 6); packed LUT (p. 7); worked model values (p. 8); lesson prototype (p. 9); implementation criteria (p. 10); provenance (p. 11); and author-identification fields (p. 12).

S2's Preparaat 1 and Preparaat 2 are represented through S2's own descriptions and equations. Their filenames and the repository names inside S1 remain part of those documents' reference layers.

Formalization method

The source sequence is reorganized into axioms, named profiles and typed notation. Explicit equations are retained. $\mathcal{K}$, $\mathcal{C}$, M_y , τ_n and $\mathfrak{P}$ are compact records of the source's already named structures. Subscripts distinguish conflicting symbol uses; $\rightsquigarrow$ records correspondences without turning them into an equals sign.

Paradigm attribution

Tom Klootwijk / NL200678942 / 10-07-1990

Document

Maktub: Formal specification of the paradigm

TK-MAKTUB-2026-01 / Version 1.0 / 09 October 2026

Source location S1 and S2 are the two supplied source documents. Page locators throughout refer to their PDF page numbers.